

ZHIXUAN LIU

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EDUCATION

Shanghai Jiao Tong University, Ph.D. in Computer Science 2022–2027

School of Electronic Information and Electrical Engineering; Advisor: Yu Qiao; expected Jun. 2027

Shanghai Jiao Tong University, B.Eng. in CS (ACM Class, Zhiyuan College) 2018–2022

RESEARCH EXPERIENCE

Shanghai Artificial Intelligence Laboratory, Research Intern 2022–Present

Subliminal Learning as Trait-Direction Drift First Author · Ongoing Work

- Introduced trait-direction drift and probe-space corridor regularization, reducing malicious-response transfer from **29.55%** to **6.47%** while preserving task performance.

Decoupled Contrastive Decoding via Expert-Aligned Drafting First Author · arXiv 2026

- Replaced the serial amateur-model proposal stage with an expert-aligned drafter while preserving exact contrastive decoding, yielding **1.65–1.95×** speedups at 8B and about **2×** at 70B.

Inference-Time Alignment via Integrated Value Guidance (IVG) First Author · EMNLP 2024

- Combined token-level implicit value guidance with chunk-level search, raising AlpacaEval 2.0 length-controlled win rates from **19.51%** / **25.58%** to **26.51%** / **33.75%**.

Weak-to-Strong Search Second Author · Core Contributor · NeurIPS 2024

- Guided chunk-level search with small-model preferences, improving Llama-3-70B from **34.42%** to **39.09%** and black-box GPT-3.5 from **16.00%** to **20.07%**.

ByteDance AI Lab, Research Intern Sep. 2020–Jun. 2021

A Neural-Symbolic Approach to Natural Language Understanding First Author · EMNLP 2022

- Proposed a neural-symbolic processor, created **32,011** DROP and **722** AWPNI program annotations, and achieved **84.01 F1** on DROP and **92.24%** accuracy on AWPNI.

OTHER PUBLICATIONS

- *Native Reasoning Models: Training Language Models to Reason on Unverifiable Data*, ICLR 2026, Second Author.
- *Adaptive Prompt Optimization for Open-Ended Tasks: Uncertainty Preference as a Secondary Signal*, ACL 2026, Second Author.
- *Scaling Model-Generated Distillation Data Can Make Latent Teacher Traits More Recoverable*, Second Author, Ongoing Work.
- *Emergent Response Planning in LLMs*, ICML 2025, Third Author.
- *SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law*, arXiv 2025, PVM Guidance Contributor.

SKILLS & HONORS

- Python, PyTorch, HuggingFace Transformers, SGLang, SFT/DPO, reward/value modeling, speculative decoding, contrastive decoding, search-based generation.
- Silver Medal, National Olympiad in Informatics (NOI), 2017.